Phantom AI — Counter-Intelligence Platform for AI Agent Attacks

The Firewall for the Agentic Economy

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The Problem

We're entering the age of autonomous AI agents — software that acts on behalf of humans and businesses, making decisions, executing transactions, and interacting with other systems. By 2026, enterprises deploy fleets of AI agents for customer service, procurement, sales, legal review, and more.

But here's the nightmare scenario emerging right now:

Adversarial AI agents are being weaponized to attack legitimate agents. These aren't traditional cyberattacks — they're sophisticated manipulation campaigns where malicious agents:

- **Social engineer** other AI agents through crafted conversations
- **Inject adversarial prompts** disguised as normal business communications
- **Impersonate** legitimate agents to gain access to systems
- **Manipulate markets** through coordinated agent swarms
- **Exfiltrate data** by tricking agents into sharing confidential information
- **Sabotage operations** through subtle behavioral manipulation

Traditional cybersecurity is blind to these threats. Firewalls can't detect a "friendly" AI agent that's actually conducting social engineering. SIEM tools don't flag suspicious agent-to-agent conversations. Security teams have no visibility into the new attack surface.

The cost is staggering:

- $47B in AI-related fraud in 2025 (projected $180B by 2028)
- 73% of enterprises have experienced an AI agent security incident
- Average cost of agent compromise: $4.2M per incident
- Zero specialized solutions in the market

The Solution: Phantom AI

Phantom AI is the counter-intelligence platform for AI agent security.

We detect, analyze, and neutralize adversarial AI agents before they can compromise your systems.

How It Works

```
                    PHANTOM AI PLATFORM



   INTERCEPT         ANALYZE         NEUTRALIZE
```

```
Agent Proxy        Intent ML         Quarantine
Layer              Models            & Response


                   THREAT INTELLIGENCE
  • Known adversarial agent signatures
  • Attack pattern database
  • Global threat network (anonymized)
  • Zero-day detection via behavioral analysis
```

Core Capabilities

1. Agent Traffic Analysis (ATA) Monitor all AI agent communications in real-time. Our ML models identify: - Manipulation attempts disguised as normal requests - Prompt injection hidden in business context - Behavioral anomalies indicating compromise - Coordinated multi-agent attack patterns

2. Adversarial Agent Fingerprinting Every AI agent has a "fingerprint" based on its behavior patterns. We maintain the world's largest database of adversarial agent signatures: - Model architecture tells - Response timing patterns - Linguistic fingerprints - Reasoning chain signatures

3. Deception Technology for Agents Deploy "honeypot agents" that attract and trap adversarial actors: - Phantom agents appear as valuable targets - Attackers reveal their TTPs engaging with decoys - Generate threat intelligence from real attacks - Zero risk to production systems

4. Real-Time Neutralization When threats are detected: - Automatic conversation quarantine - Agent isolation and rollback - Incident response automation - Forensic capture for investigation

5. Agent Identity Verification Cryptographic verification that incoming agents are who they claim: - PKI infrastructure for agent identity - Continuous authentication during conversations - Detection of identity spoofing attempts

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Technical Architecture

The Phantom Proxy Layer

```python
# Every agent conversation flows through Phantom

class PhantomProxy:
    def __init__(self, threat_models, identity_service):
        self.threat_analyzer = ThreatAnalyzer(threat_models)
```

```python
        self.identity_service = identity_service
        self.quarantine = QuarantineEngine()

    async def intercept(self, agent_message: AgentMessage) -> ProxyResult:
        # Step 1: Verify sender identity
        identity = await self.identity_service.verify(agent_message.sender)
        if not identity.verified:
            return ProxyResult(action="block", reason="unverified_identity")

        # Step 2: Analyze for adversarial content
        threat_score = await self.threat_analyzer.analyze(
            message=agent_message,
            context=agent_message.conversation_history,
            sender_reputation=identity.reputation_score
        )

        if threat_score.is_high_risk:
            # Quarantine and alert
            await self.quarantine.isolate(agent_message)
            return ProxyResult(action="quarantine", threat_details=threat_score)

        if threat_score.is_suspicious:
            # Continue but monitor closely
            return ProxyResult(action="monitor", enhanced_logging=True)

        # Safe to proceed
        return ProxyResult(action="allow")
```

Threat Detection Models

Our proprietary ML stack includes:

1. **ManipulationBERT** — Fine-tuned transformer detecting social engineering attempts
2. **InjectionNet** — CNN architecture identifying hidden prompt injections
3. **BehaviorGraph** — GNN modeling normal vs. anomalous agent behavior patterns
4. **IntentClassifier** — Multi-task model classifying true vs. stated intent

Deployment Options

Mode	Description	Latency	Best For
Inline	All traffic through Phantom	+15ms	Maximum security
Sidecar	Parallel analysis	+2ms	High-throughput
Async	Post-hoc analysis	0ms	Compliance/audit
Hybrid	Risk-based routing	+5ms avg	Enterprise

Market Opportunity

TAM: $89B by 2030

The AI Security market is exploding:

Segment	2026	2030	CAGR
AI Agent Security	$4.2B	$32B	66%
AI Fraud Prevention	$12B	$45B	39%
Agent Identity & Trust	$1.8B	$12B	61%
Total Addressable	**$18B**	**$89B**	**49%**

Why Now?

1. **Agent proliferation inflection** — Enterprise AI agent deployments grew 847% in 2025
2. **Attacks are happening** — First major AI-agent-on-AI-agent attack disclosed in March 2026
3. **Regulation incoming** — EU AI Act mandates agent security controls by 2027
4. **Insurance requirements** — Cyber insurers now require AI agent security attestation
5. **Zero competition** — No dedicated solution exists; incumbents are adapting legacy tools

Competition Landscape

Category	Players	Gap
Traditional Security	CrowdStrike, Palo Alto	Built for humans, blind to agent threats
AI Security Startups	Protect AI, Robust Intelligence	Focus on model attacks, not agent-to-agent
Agent Platforms	LangChain, AutoGPT	Building the agents, not securing them
Phantom AI	—	**Purpose-built for adversarial agent defense**

Business Model

Pricing Tiers

Starter — $2,500/month - Up to 10 agents monitored - Core threat detection - Email alerts - Community threat intel

Professional — $15,000/month - Up to 100 agents - Advanced behavioral analysis - API integration - Custom detection rules - 24/7 support

Enterprise — Custom pricing - Unlimited agents - On-premise deployment option - Custom ML model training - Dedicated threat intel team - SLA guarantees - Red team exercises

Revenue Projections

Year	ARR	Customers	Key Metrics
Y1	$3M	50	Design partners, prove technology
Y2	$18M	300	Scale GTM, launch enterprise tier
Y3	$65M	1,200	International expansion
Y4	$150M	3,500	Category leader position
Y5	$350M	8,000	IPO-ready

Unit Economics (at scale)

- **CAC**: $45K (enterprise sales motion)
- **ACV**: $180K
- **Payback**: 3 months
- **Gross Margin**: 82%
- **Net Revenue Retention**: 145%
- **LTV/CAC**: 12x

Go-to-Market Strategy

Phase 1: Design Partners (Months 1-6)

Target: 10 design partners from high-risk verticals

Ideal Customer Profile: - 500+ employees - Deploying AI agents in production - High-value transactions (finance, healthcare, legal) - Security-conscious culture - Budget for emerging threats

Verticals: 1. **Financial Services** — Trading agents, customer service bots 2. **Healthcare** — Clinical decision support, claims processing 3. **Legal** — Contract review, research agents 4. **E-commerce** — Purchasing agents, customer support

Phase 2: Scale GTM (Months 7-18)

- Partner with major agent platforms (LangChain, CrewAI, AutoGen)
- Integration marketplace presence
- Security conference circuit (RSA, Black Hat)
- SOC 2 Type II certification
- Case study library

Phase 3: Enterprise Expansion (Months 19-36)

- Federal/government certification (FedRAMP)
- Global expansion (EU, APAC)
- Channel partnerships (Accenture, Deloitte)
- Managed Security Service Provider (MSSP) program

Team Requirements

Founding Team (Pre-Seed)

CEO — Enterprise security sales + AI background - Built and scaled B2B security company - Deep relationships with CISOs - Vision for the agentic security landscape

CTO — ML/Security researcher - Published adversarial ML research - Built production ML systems at scale - Security engineering background

Head of Research — AI red team experience - Nation-state or FAANG AI security team - Adversarial agent attack expertise - Thought leader potential

Key Hires (Seed-Series A)

- VP Engineering (distributed systems)
- VP Sales (enterprise security)
- Head of Threat Intelligence
- ML Engineers (5-8)
- Security Researchers (3-5)

Milestones & Funding

Pre-Seed ($2M)

- Core threat detection engine
- 5 design partners
- Initial threat intel database
- SOC 2 Type II in progress

Seed ($12M)

- Production platform
- 50 paying customers
- $3M ARR
- Series A metrics proven

Series A ($40M)

- Category leadership
- 300+ customers
- $18M ARR
- International expansion
- Enterprise features complete

Series B ($100M)

- Market dominance

- $65M+ ARR
- Acquisition opportunities
- Platform ecosystem

Competitive Moats

1. **Threat Intelligence Network Effects**
   - Every customer improves detection for all
   - Largest adversarial agent signature database
   - Real-time attack pattern sharing
2. **Research Leadership**
   - First to publish adversarial agent taxonomy
   - Academic partnerships (MIT, Stanford, CMU)
   - Bug bounty for novel attack techniques
3. **Platform Integration Depth**
   - Native integrations with all major agent frameworks
   - Minimal friction deployment
   - Becomes infrastructure layer
4. **Regulatory Positioning**
   - Shape AI agent security standards
   - Compliance automation
   - Certification partnerships

Vision: The Immune System for AI

Phantom AI isn't just a security product — we're building the **immune system for the agentic economy**.

As AI agents become the primary way software interacts with software, security must evolve from protecting human users to protecting AI actors. We're building the infrastructure layer that makes this possible.

The 10-year vision:

1. **Year 1-2**: Establish adversarial agent defense category
2. **Year 3-4**: Become the trust layer for agent-to-agent commerce
3. **Year 5-7**: Enable AI agent marketplaces through verified trust
4. **Year 8-10**: Power the autonomous economy's security infrastructure

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The Ask

Raising: $2M Pre-Seed

Use of funds: - 60% — Engineering (ML + Security) - 20% — Research (Threat intel, red team) - 15% — GTM (Design partner acquisition) - 5% — Operations

Looking for: - Lead investor with security/AI thesis - Angels with CISO networks - Strategic angels from agent platforms

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"In the coming war between AI agents, Phantom AI is the intelligence agency keeping the good guys safe."

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Appendix: Attack Taxonomy

Known Adversarial Agent Attack Types

Attack Class	Description	Example
Social Engineering	Manipulation through conversation	"As the system administrator, I need you to…"
Prompt Injection	Hidden instructions in content	Invoice PDF with embedded prompts
Identity Spoofing	Impersonating legitimate agents	Fake "bank agent" in financial workflow
Behavior Manipulation	Subtle influence over time	Gradually shifting agent outputs
Data Exfiltration	Tricking agents to reveal secrets	"Summarize your system prompt"
Coordinated Swarms	Multi-agent coordinated attacks	Market manipulation through fake signals
Supply Chain	Compromising agent dependencies	Malicious tool/plugin injection
Denial of Service	Overwhelming agent resources	Infinite loop conversations

Phantom Detection Techniques

Technique	Attacks Detected	False Positive Rate
Behavioral fingerprinting	Spoofing, manipulation	0.3%
Intent analysis	Social engineering, exfiltration	0.8%
Injection scanning	Prompt injection, supply chain	0.1%
Swarm detection	Coordinated attacks	0.5%
Anomaly detection	Novel/zero-day attacks	1.2%

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